# Curriculum Vitae

**Jonas Köppl**
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## Academic appointments

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|---|---|
| 10/2026 – | Postdoctoral researcher, Universität Münster. |
| 04/2026 – 09/2026 | Research assistant, Technische Universität Braunschweig. |
| 09/2022 – 03/2026 | Research assistant, Weierstrass Institut Berlin. |

## Academic education

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|---|---|
| 09/2022 – 07/2026 | Dr. rer. nat. (*summa cum laude*), Technische Universität Berlin. Supervisor: Benedikt Jahnel. |
| 01/2025 – 06/2025 | Visiting graduate student, University of California, Los Angeles. Mentors: Marek Biskup and Georg Menz. |
| 10/2019 – 03/2022 | M.Sc. Mathematics, Technische Universität Berlin. |
| 10/2016 – 08/2019 | B.Sc. Mathematics, Universität Passau. |

## Research interests

Probability and statistical mechanics with a focus on the long-term behaviour of interacting particle systems and phase transitions in random graphs and complex networks.

## Selected publications

**Jonas Köppl** and Yannic Steenbeck. *Asymptotically complete free-energy dissipation: a coarse MLSI holds at any positive temperature*. arXiv:2608.13677 (2026).

**Jonas Köppl**. *On the absence of time-translation symmetry breaking in some non-reversible interacting particle systems*. arXiv:2602.22104 (2026).

David Coupier, Benoît Henry, Benedikt Jahnel, and **Jonas Köppl**. *The planar lattice two-neighbor graph percolates*. Accepted for publication in Annals of Probability; arXiv:2412.20781 (2024).

Benedikt Jahnel and **Jonas Köppl**. *On the long-time behaviour of reversible interacting particle systems in one and two dimensions*. Probability and Mathematical Physics 6(2), 479–503 (2025).

Benedikt Jahnel and **Jonas Köppl**. *Dynamical Gibbs variational principles for irreversible interacting particle systems with applications to attractor properties*. Annals of Applied Probability 33(6A), 4570–4607 (2023).

## Awards and scholarships

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|---|---|
| 09/2025 | Oberwolfach Leibniz Graduate Student (travel grant). |
| 01/2025 – 06/2025 | Research scholarship, DAAD (German Academic Exchange Service). |
| 09/2023 | Participant, 10th Heidelberg Laureate Forum 2023. |
| 11/2022 | Best talk award, Dies Mathematicus 2022, TU Berlin. |
| 08/2022 | Best talk award, DMV Student Conference 2022, MPI MiS Leipzig. |
| 10/2018 – 03/2022 | Scholarship holder, Studienstiftung des deutschen Volkes. |
| 01/2020 | Fakultätspreis, Fakultät für Informatik und Mathematik, Universität Passau. |

## Thesis supervision

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| 2024 – 2025 | Thomas Worschech (Master): *All stationary measures for interacting Ornstein–Uhlenbeck processes on regular graphs are reversible* (jointly with J.-D. Deuschel, TU Berlin). |

**Organisation of scientific events**

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| Since 05/2024 | Co-organiser of the monthly *Postgraduate Online Probability Seminar* (POPS). |
| 07/2026 | Co-organiser of the 3-day workshop *Spreading dynamics in random environments*, WIAS. |
| 11/2024 | Co-organiser of the 3-day workshop *Probability meets biology*, WIAS. |
| 08/2024 | Organiser of the contributed session *Gibbs measures: continuum and discrete*,<br>IMS Bernoulli World Congress 2024, Bochum. |
| 01/2024 | Co-organiser of the 3-day workshop *Dynamic spatial random systems*, WIAS. |

**Reviewing activities**

Electronic Journal of Probability, Stochastic Processes and their Applications, Annales de l'Institut Henri Poincaré – Probabilités et Statistiques.

**Research stays**

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|---|---|
| 09/2026 – 10/2026 | Erwin Schrödinger Institut Wien. |
| 01/2026 | INRIA Lille (David Coupier and Benoît Henry). |
| 11/2025 | University of California Los Angeles (Marek Biskup and Johannes Bäumler). |
| 01/2025 – 06/2025 | University of California Los Angeles (Marek Biskup and Georg Menz). |
| 03/2024 | Arizona State University (Nicolas Lanchier). |
| 10/2022 | University of New South Wales (Gary Froyland). |

# Publication list

## Published

[1] Nataša Djurdjevac Conrad, **Jonas Köppl**, and Ana Djurdjevac. *Feedback loops in opinion dynamics of agent-based models with multiplicative noise.* Entropy 24(10), article 1352 (2022).

[2] Benedikt Jahnel and **Jonas Köppl**. *Trajectorial dissipation of $\Phi$-entropies for interacting particle systems.* Journal of Statistical Physics 190, article 119 (2023).

[3] Benedikt Jahnel and **Jonas Köppl**. *Dynamical Gibbs variational principles for irreversible interacting particle systems with applications to attractor properties.* Annals of Applied Probability 33(6A), 4570–4607 (2023).

[4] Benedikt Jahnel and **Jonas Köppl**. *On the long-time behaviour of reversible interacting particle systems in one and two dimensions.* Probability and Mathematical Physics 6(2), 479–503 (2025).

[5] **Jonas Köppl**, Nicolas Lanchier, and Max Mercer. *Survival and extinction for a contact process with a density-dependent birth rate.* Electronic Journal of Probability 30, article 74, 1–18 (2025).

[6] Benedikt Jahnel, **Jonas Köppl**, Bas Lodewijks, and Andras Tobias. *Percolation in lattice $k$-neighbor graphs.* Journal of Applied Probability 62(4) (2025).

[7] Benedikt Jahnel and **Jonas Köppl**. *Time-periodic behaviour in one- and two-dimensional interacting particle systems.* Annales Henri Poincaré (online first, 2025).

[8] Benedikt Jahnel, **Jonas Köppl**, Yannic Steenbeck, and Alexander Zass. *The variational principle for a marked Gibbs point process with infinite-range multibody interactions.* Electronic Journal of Probability 30, article 176, 1–32 (2025).

[9] Edward Athaide, Maciej Głuchowski, **Jonas Köppl**, and Georg Menz. *A correlation bound for the one-dimensional heterogeneous Ising model.* Electronic Communications in Probability 31, 1–13 (2026).

## Accepted

[10] David Coupier, Benoit Henry, Benedikt Jahnel, and **Jonas Köppl**. *The planar lattice two-neighbor graph percolates.* Accepted for publication in Annals of Probability; arXiv:2412.20781 (2024).

## Preprints

[11] **Jonas Köppl**, Nicolas Lanchier, and Max Mercer. *Evolutionary games on the lattice: multitype contact process with density-dependent birth rates.* arXiv:2412.19957 (2024).

[12] Benedikt Jahnel, **Jonas Köppl**, Yannic Steenbeck, and Alexander Zass. *Reversible birth-and-death dynamics in continuum: free-energy dissipation and attractor properties.* arXiv:2508.21196 (2025).

[13] Johannes Bäumler, Benedikt Jahnel, **Jonas Köppl**, Bas Lodewijks, Lily Reeves, and Andras Tobias. *Local criteria for global connectivity comparisons: beyond stochastic domination.* arXiv:2510.03934 (2025).

[14] Benedikt Jahnel, **Jonas Köppl**, Yannic Steenbeck, and Alexander Zass. *Reversible birth-and-death dynamics in continuum: a de Bruijn-type identity for free-energy dissipation.* arXiv:2602.13474 (2026).

[15] **Jonas Köppl**. *On the absence of time-translation symmetry breaking in some non-reversible interacting particle systems.* arXiv:2602.22104 (2026).

[16] Benedikt Jahnel and **Jonas Köppl**. *Restriction and mixing properties of interacting particle systems with unbounded range.* arXiv:2603.21817 (2026).

[17] Benedikt Jahnel, **Jonas Köppl**, Lukas Lüchtrath, and Anh Duc Vu. *Convex order and faster transmission in first contact percolation.* arXiv:2605.28766 (2026).

[18] **Jonas Köppl** and Yannic Steenbeck. *Asymptotically complete free-energy dissipation: a coarse MLSI holds at any positive temperature.* arXiv:2608.13677 (2026).

[19] Jean-Dominique Deuschel, **Jonas Köppl**, Yannic Steenbeck, and Thomas Worschech. *On the stationary measures of the critical Ornstein–Uhlenbeck process.* arXiv:2609.07622 (2026).

[20] Benedikt Jahnel, **Jonas Köppl**, and Peteris Siliņš. *Comparison inequalities for discrete- and continuous-time infection processes.* arXiv:2609.09892 (2026).

## Talks

**2026**

Technische Universität Darmstadt, *Oberseminar Stochastik*, July 2026, Darmstadt.
Technische Universität Braunschweig, *Oberseminar Stochastik*, April 2026, Braunschweig.
University of Maryland, *Statistical Mechanics of Lattice Systems*, March 2026, College Park.
Université de Lille, Groupe de travail "*Processus ponctuels et applications*", January 2026, Lille.

**2025**

Universität Ulm, *Oberseminar für angewandte Analysis*, November 2025, Ulm.
MFO Oberwolfach, *Large scale stochastic dynamics*, September 2025, Oberwolfach.
Stochastic Processes and Applications, July 2025, Wrocław.
Caltech, LA Probability Forum, March 2025, Los Angeles.
UCLA, Probability Seminar, February 2025, Los Angeles.

**2024**

EURANDOM, Interacting particles in the continuum, September 2024, Eindhoven.
4th Italian Meeting on Probability and Mathematical Statistics, June 2024, Rome.
Universität Paderborn, Probability seminar, May 2024, Paderborn.
Postgraduate Online Probability Seminar (POPS), February 2024, online.
UCLA, Probability seminar, February 2024, online.

**2023**

Mathematics of random systems summer school, September 2023, Kyoto.
German Probability and Statistic Days 2023, March 2023, Essen.
BMS Student Conference 2023, February 2023, Berlin.
Universität Passau, *Analysis and Probability* Seminar, January 2023, Passau.

**2022**

TU Berlin, Dies Mathematicus, November 2022, Berlin.
WIAS *Interacting Random Systems* Seminar, November 2022, Berlin.
DMV Annual Meeting, September 2022, Berlin.
BMS–BGSMath Junior Meeting, September 2022, Barcelona.
DMV Student Conference, August 2022, Leipzig.